

Robert Garabetian

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EDUCATION

University of Southern California

Los Angeles, CA

Bachelor of Science – Computer Science

Graduating December 2027

- **Relevant Coursework:** Object-Oriented Programming, Probability Theory, Advanced Python, Data Structures, Embedded Systems, Algorithms, Software Development, Computer Systems, Professional C++

EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Layup Parts (Full-time)

Huntington Beach, CA

- Engineered a production automation system using **Go**, **React**, and **TypeScript** that integrated the company's internal factory software with an XTool laser engraver, automating serial number allocation and engraving for carbon fiber drone arms and increasing manufacturing throughput by **250%** while reducing operator error by **95%**.
- Built a factory-wide **CI/CD** deployment pipeline using **Python**, **GitLab CI**, and the **Portainer API** that automated updates for **20 Docker containers** running on **Raspberry Pi** barcode scanner systems, eliminating manual deployments and significantly reducing maintenance effort.
- Designed an engineering operations dashboard in **Go** backed by a **Neo4j** graph database that unified data from the company's client-facing software and factory systems, enabling manufacturing leadership to allocate engineers, track project lifecycles, and surface operational insights.
- Built a go/links URL shortener with custom **DNS** settings, letting all office staff reach shared resources through customizable short links like go/teams.
- Developed a production documentation system integrating **AWS S3** and **Neo4j**, enabling technicians and quality staff to attach notes and photo evidence to manufacturing operations, with a microservice that standardized uploaded image formats.

Director of Curriculum

Jan 2026 – May 2026

Scope (USC) (Part-time)

Los Angeles, CA

- Designed and taught a project-based **Docker** and **Kubernetes** curriculum to 30+ students, emphasizing real-world deployment and container orchestration.

Software Engineer Intern

Apr 2025 – Dec 2025

MGD Consulting (Part-time)

Glendale, CA

- Developed dashboards using **TypeScript** and **Next.js** that surfaced medical coding and patient plan of care status, supporting operations for **250+ home health agencies** managing thousands of patients.
- Collaborated with senior developers to introduce backend service deployment infrastructure to **AWS** using **Docker** and **Kubernetes**, reducing time-to-production by **40%** through automated **CI/CD** pipelines.

Coding Instructor

Oct 2023 – Sep 2024

Technix Academy (Part-time)

Montrose, CA

- Delivered structured lessons to **80+ students**, teaching foundational through college-level computer science concepts including object-oriented programming and algorithmic thinking using **Python**, **Java**, and **C++**.
- Led hands-on classes, introducing students to digital tools such as Adobe Illustrator and CAD.
- Collaborated with supervisors to develop and revise over **200 pages** of **C++** and **Python** curriculum for teaching students Data Structures and Algorithms across multiple structured classroom learning modules.

PROJECTS

Video Transcoder

- Engineered a video transcoding pipeline using **Go**, **FFmpeg**, and the **AWS SDK** to convert video uploads into adaptive **HLS** with concurrent cleanup routines using **goroutines** to automatically delete temporary local files after uploads, and to periodically purge stale files from the **S3** bucket to reduce disk usage.

TECHNICAL SKILLS

Languages: C++, Go, TypeScript, JavaScript, Python, Java, SQL, Cypher, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, REST APIs

Databases & Cloud: PostgreSQL, MySQL, Neo4j, Supabase, AWS, GCP, DigitalOcean, Vercel, Docker, Kubernetes

Developer Tools: Git, GitHub, GitLab, Linux, VS Code, Cursor, Claude Code, Codex